

EngiAI — Post-Thesis Work Plan

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Project: EngiAI — A Multi-Agent Framework and Benchmark Suite for LLM-Driven Engineering Design

Goals

- 1. Make the system production-ready and easy to deploy** — Ensure full test coverage, update all dependencies, simplify the Docker-based deployment with a graphical launcher for non-technical users, standardize the codebase to match the lab's EngiBench/EngiOpt conventions, and set up automated CI/CD pipelines.
- 2. Extend the system with new capabilities** — Add support for local language models (Ollama) in the chat interface, develop an interactive whiteboard for collaborative design (Excalidraw integration), implement a new agent for automated training code generation with HPC support, extend the document retrieval system to additional file formats, set up experiment recording infrastructure, improve error handling, and configure a new dedicated workstation.

Key Milestones

Date	Event
Apr 3–6, 2026	Easter break
Apr 13, 2026	Thesis written report deadline
Apr 20–23, 2026	HPC AI conference (tentative)
Apr 27, 2026	IDETC paper notification
May 8, 2026	Final presentation (30 min + questions)
May 19, 2026	IDETC final paper submission + author registration
Jun 29 – Jul 7, 2026	PASC conference + holidays
Aug 23–26, 2026	IDETC conference

Swiss public holidays (Labour Day, Ascension, Whit Monday) are accounted for in the schedule below. All hour estimates are approximate — actual effort may vary depending on task complexity and external factors.

Phase 1: Thesis Completion + Demo Video + Code Quality (Mar 17 – Apr 13) — ~125h

Period	Task	Hours
Mar 17–20	Thesis polishing + YouTube video script	25h
Mar 23–27	Record and edit demo video (5–8 min, all system features)	30h
Mar 30 – Apr 2	Fix test suite + update dependencies + Python 3.12 support	30h
Apr 7–10	Start presentation slides	30h
Apr 13	Final thesis compilation and submission	10h

Deliverables: Thesis submitted. Demo video published. Test suite fully passing. Dependencies up to date. Presentation draft started.

Phase 2: Presentation + Standardization + arXiv Preprint (Apr 14 – May 8) — ~110h

Period	Task	Hours
Apr 14–17	Finalize presentation + rename repository to EngiAI + deployment simplification (Docker profiles, secure credentials, dependency management)	30h
Apr 20–23	HPC AI conference (unavailable)	0h
Apr 24	Presentation refinement	10h
Apr 27–30	Codebase standardization (EngiBench/EngiOpt conventions, CI/CD pipelines, documentation update) + rehearsal	30h
May 4–7	arXiv preprint preparation + rehearsals + begin GUI launcher development	30h
May 8	Final presentation	10h

Deliverables: Presentation delivered. arXiv preprint submitted. Codebase standardized with automated CI/CD. GUI launcher prototype with secure settings management.

Phase 3: Paper Revision + Code Cleanup + System Improvements (May 9 – Jun 27) — ~110h

3a. IDETC Paper Revision (May 9–19) — 20h

Address reviewer feedback (if needed), prepare camera-ready version, and complete author registration by the **May 19 deadline**. Finalize the GUI launcher in parallel.

3b. Code Cleanup (May 19 – Jun 6) — 35h

- Remove dead code, duplicate logic, and unused dependencies
- Standardize the Prusa 3D printer integration module to lab conventions
- Extract the Excalidraw whiteboard component into a standalone reusable package

3c. System Improvements (Jun 8–26) — 55h

Period	Task	Hours
Jun 8–12	Local model support (Ollama integration in chat interface)	15h
Jun 15–19	Improved progress feedback for long-running tasks + error recovery mechanisms	20h
Jun 22–26	Experiment recording infrastructure (screen + camera capture) + Google audio model support	20h

Jun 29 – Jul 7: PASC conference + holidays (unavailable)

Phase 4: New Features + Handover (Jul 8 – Aug 7) — ~105h

4a. Core Features (Jul 8–24) — 55h

Period	Task	Hours
Jul 8–17	Interactive Excalidraw whiteboard with AI agent integration (bidirectional: agent draws results, user sketches constraints)	35h
Jul 20–24	Additional file format support for document retrieval + containerized problem solver framework	20h

4b. New Agent + Handover (Jul 27 – Aug 7) — 50h

Period	Task	Hours
Jul 27 – Aug 3	Training code generation agent (automated code validation, HPC integration for Euler, full test suite)	30h

Period	Task	Hours
Aug 4–5	Repository transfer to lab GitHub organization (v2.0.0 release)	10h
Aug 6–7	New workstation setup and deployment verification	10h

Phase 5: IDETC Conference + Wrap-Up (Aug 10–28) — ~60h

Period	Task	Hours
Aug 10–21	IDETC conference presentation preparation, rehearsals + travel	30h
Aug 23–26	IDETC conference	20h
Aug 27–28	Final documentation, release tagging, status report for the lab	10h

Summary

Phase	Period	Pace	Hours
1. Thesis + Video + Code Quality	Mar 17 – Apr 13	Full-time	~125h
2. Presentation + Standardization	Apr 14 – May 8	Full-time	~110h
3. Paper Revision + Cleanup + Improvements	May 9 – Jun 27	Part-time	~110h
4. New Features + Handover	Jul 8 – Aug 7	Part-time	~105h
5. IDETC + Wrap-Up	Aug 10 – Aug 28	Part-time	~60h
Total	Mar 17 – Aug 28		~510h

The full-time period (Phases 1–2) covers all deadline-critical work: thesis, presentation, arXiv preprint, and codebase standardization. The part-time period (Phases 3–4) is modular — tasks can be reordered or adjusted depending on the IDETC paper revision workload.

Contingencies

Scenario	Response
Heavy IDETC paper revision required	Defer some Phase 3b cleanup tasks to Phase 4 buffer
IDETC paper accepted without revision	Redirect ~30h to feature development

Scenario	Response
Excalidraw integration more complex than estimated	Deliver a simplified version and document the full design

Stretch Goals

If buffer hours are available:

- **Ablation studies** (prompt/temperature/model comparisons) — ~15–20h. Useful for potential future publications.

Proposed Employment (tentative)

Employment period	~May 9 – August 28, 2026 (part-time, after thesis defense)
Employment rate	~50% (~2 full days + 1 half day per week, ~20h/week)
Total part-time effort	~275 hours across ~13 usable weeks

Exact dates and employment rate to be discussed. Prior to the employment period, Phases 1–2 (Mar 17 – May 8) are covered under the existing full-time arrangement (~40h/week, ~235h). The part-time contract would begin after the final presentation on May 8.

Total project effort: ~510 hours (full-time + part-time combined).